

STEMS: Safe Multi-Agent Reinforcement Learning for Buildings, Heat Pumps and Electric Vehicles

From replicating a paper to a calibrated, anticipatory safety layer:
every idea, state, equation and result

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- 1 Overview and chain of reasoning
- 2 Starting point: replicating STEMS
- 3 The environment
- 4 The states: what the agent sees and where it comes from
- 5 The learning architecture
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The project in one slide

Problem

Coordinate the **battery**, **heat pump**, **hot-water tank** and **EV charger** of several houses, while respecting

- device limits (state of charge, power),
- occupant comfort and hot-water readiness,
- EV departure requirements,
- a **shared grid connection**.

What we built

A reimplementaion of STEMS on real CityLearn data, a calibrated and anticipatory safety layer, one barrier for hot water, EVs and batteries, a buildings + EV environment, and an evaluation harness.

What we learned

- ① Actuator dynamics must be **measured from the plant**, not assumed.
- ② Anticipation horizons must match **measured lead times**.
- ③ Coupled deadlines under a shared cap can make the **safe set empty**.
- ④ The value of coordination is **non-monotone** in how tightly the cap binds.
- ⑤ A shield that guarantees an objective **removes the learning signal** for it.

Where we are

Clean evaluation around one research question (next slide).

The research question

Main question (agreed with Andreas)

How much does a **calibrated and anticipatory safety layer** improve **constraint satisfaction** in RL-based multi-device energy management, and what is the resulting **cost / comfort trade-off** under increasingly constrained operating conditions?

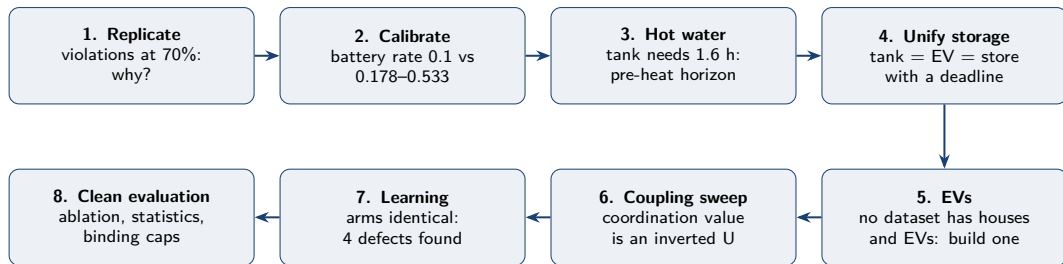
Sub-questions

- Does the improvement come from the **policy**, the **barrier**, or both?
- When does **coordination** between houses matter?
- How robust is it to **forecast errors**?

Scope

- Devices: battery, heat pump, hot-water tank, EV.
- Environments: Texas (cooling), then a heating-dominated scenario, then Thermia data.
- Standalone heat-pump mode must stay reproducible (Manal's benchmark).

The journey: each step answered a question raised by the previous one



Pattern at every step: **question** → **measurement** → **mechanism** → **experiment** → **what broke** → next question.

Working principles, and the failure behind each

Principle	Why we adopted it
Measured, not assumed	A single hard-coded battery rate (0.1) produced most constraint violations. Every rate, capacity and efficiency is now read from the simulator.
Fail loud	The original code silently swapped in a mock environment and emitted zero actions when infeasible. Missing data, devices or observations now raise; simulator patches print a banner and are stamped into run meta-data.
Causal forecasts only	Hot-water demand is forecast from already-observed steps; nothing reads the future.
Report negative results	Pre-heating costs money; an inactive mechanism is shown as inactive; each comes with a planned remedy.
Regression-check every refactor	The heat-pump-only results are a benchmark in Manal's PhD: every structural change is diffed against the stored results.
Verify actuators in every run	The hot-water actuator was inert for months without any error. Each run now proves its actuators respond.

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Zhang, Wu, Zinflou and Boulet (2025), IEEE Internet of Things Journal, arXiv:2510.14112.

- Multi-agent safe RL for building energy management on CityLearn.
- **Spatial** encoder: graph convolution over a building-similarity graph.
- **Temporal** encoder: self-attention over a 24 h history.
- Per-building **actor-critic** (SAC-style) with **cost critics**.
- **Control barrier function** shield on battery state of charge and power.
- **Lagrangian** pressure so the policy itself learns to be safe.

Our goal at the start

Reimplement it faithfully on **real** data, with every modelling assumption checked against the simulator.

What we found in the original code

Silent fallbacks

- Environment silently swapped to a synthetic **mock** when the CityLearn import failed.
- CBF returned **all-zero actions** when its optimisation was infeasible.
- Trainer used **raw, unsafe actions** when safe actions were missing.
- One **uniform battery rate** (0.1 per unit action) for every building.

Consequence

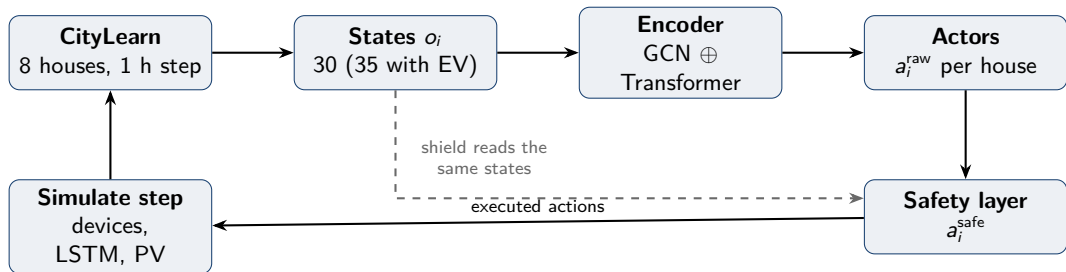
The one honest run on real data sat at about **70% constraint violations**.

Response

- Rewrite fail-loud: the mock is reachable only with an explicit flag and a banner.
- `env_type` and active mechanisms stamped into every result.
- Analytic, feasibility-guaranteed projection (never zeros).
- Per-building calibration from the simulator.

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The control loop



Every hour: observe $\rightarrow$ encode $\rightarrow$ propose $\rightarrow$ project onto the safe set $\rightarrow$ simulate. Rewards, costs and KPIs are computed from the **observed** next state, never from the shield's internal model.

Dataset and scenario

Simulator	CityLearn 2.6.0b1 (Gymnasium API)
Building data	NREL ResStock, release amy2018–2021; calendar matches 2018
Location	Travis County, Texas (cooling-dominated)
Buildings	8 simulated out of 100 candidates; decentralised control
Horizon	8760 hourly steps (one year)
Devices per house	heating heat pump, cooling heat pump, electric hot-water heater, hot-water tank, battery, rooftop PV
Building type	LSTMDynamicsBuilding (indoor temperature from an LSTM)
Device sizing	autosized by CityLearn at construction , from the simulation period
Active observations	45; STEMS selects 28 by name, 30 with heat-pump features
Actions	[dhw_storage, electrical_storage, cooling_or_heating_device]
Prices, carbon	files borrowed from the CityLearn 2022 challenge (California)

Devices and their sizes (full-year autosizing, all 100 candidates)

Device	Model	Size range
Heating	heat pump	2.04–21.10 kW
Cooling	heat pump	1.10–12.53 kW
Hot-water heater	electric heater	3.94–17.32 kW
Hot-water tank	storage tank	0–12.48 kWh
Battery	battery	3.30–24.50 kWh 1.28–7.07 kW
PV	PySAM PVWatts	3.82–26.80 kW

Not every house is equipped

- **61 of 100** candidates have **no hot-water tank**.
- **39 of 100** have every device.
- The reference eight are all fully equipped.

Our wrapper reported missing tanks as capacity 1.0: a silent fallback, **to fix**. Building subsets will be drawn from the 39 fully equipped houses (the sampler currently draws from all 100).

Device physics: storage (battery and hot-water tank)

Both devices share CityLearn's storage model. For requested energy e (positive = charge), capacity C , round-trip efficiency η , standby loss λ :

$$\text{SOC}^{\text{init}} = \max(0, \text{SOC}_{t-1} C (1 - \lambda))$$

$$\text{SOC}_t = \frac{1}{C} \begin{cases} \min(\text{SOC}^{\text{init}} + e\sqrt{\eta}, C) & e \geq 0 \quad (\text{charge}) \\ \max(0, \text{SOC}^{\text{init}} + e/\sqrt{\eta}) & e < 0 \quad (\text{discharge}) \end{cases}$$

Battery: $e = a P_{\text{nom}} \Delta t$. So the state of charge moves by about

Hot-water tank: $e = a C$, clipped to what the heater can deliver in one hour (next slides).

$$\rho_i = P_{\text{nom},i} / C_i \quad \text{per unit action.}$$

Why this matters

ρ_i is the quantity the safety barrier needs. Getting it wrong by a factor of 2–5 was the main source of violations.

Device physics: heat pump

Carnot coefficient of performance, efficiency η , supply temperature T_{tgt} (measured: $\eta \in [0.21, 0.29]$ cooling, $[0.20, 0.28]$ heating; supply 7.2–9.8 and 45.2–48.8 °C):

$$\text{CoP}_{\text{heat}}(T_{\text{out}}) = \eta \frac{T_{\text{tgt}}^{\text{heat}} + 273.15}{T_{\text{tgt}}^{\text{heat}} - T_{\text{out}}}, \quad \text{CoP}_{\text{cool}}(T_{\text{out}}) = \eta \frac{T_{\text{tgt}}^{\text{cool}} + 273.15}{T_{\text{out}} - T_{\text{tgt}}^{\text{cool}}}$$

$$P_{\text{out}} = \min(|a| P_{\text{nom}}, P_{\text{nom}}) \cdot \text{CoP}, \quad P_{\text{in}} = P_{\text{out}} / \text{CoP}, \quad \text{CoP} \leq 20$$

The sign of the action picks the mode

CityLearn splits one action into two devices, $a^{\text{cool}} = |\min(a, 0)|$ and $a^{\text{heat}} = \max(a, 0)$: $a < 0$ cools, $a > 0$ heats, each against its own nominal power. The hvac_mode gate can disable a mode, but it equals 3 (both allowed) at **every hour of all eight houses**.

Consequences

- Comfort needs a **dual set-point deadband**: the agent may heat or cool at any hour.
- Anything that assumes the mode — our CoP guard reads it off the weather, the rule-based baseline sends +0.2 “for cooling” — is wrong by construction (see the defects).

Device physics: hot water, and the lead time

Electric heater: $P_{\text{out}} = \min(|a|P_{\text{nom}}, P_{\text{nom}})\eta$, $\eta \approx 0.9\text{--}0.99$. The tank is charged by $e = aC$, clipped to the heater's hourly output, so the state-of-charge gain per step is

$$\rho_i^{\text{dhw}} = \min\left(c_i, \frac{P_{\text{nom},i}\eta_i}{C_i}\right), \quad \tau_i = \frac{1}{\rho_i^{\text{dhw}}} \text{ hours to fill an empty tank}$$

with c_i CityLearn's per-building action bound.

Measured on the reference eight

$\rho^{\text{dhw}} \in [0.52, 0.85]$, $\tau \in [1.17, 1.92]$ h,
mean 1.60 h.

Validated: one maximum-charge step from empty lands within 15% of ρ_i^{dhw} in every house.

The idea this created

Hot water cannot be made on demand. A controller that reacts **after** seeing demand is structurally too late: it must **pre-heat**, with a horizon at least as long as τ .

How net consumption, cost and emissions are computed

CityLearn, every step, for building i (PV generation is stored as a **negative** value):

$$e_{i,t} = E_{i,t}^{\text{cool}} + E_{i,t}^{\text{heat}} + E_{i,t}^{\text{dhw}} + E_{i,t}^{\text{nsf}} + E_{i,t}^{\text{batt}} + E_{i,t}^{\text{EV}} + E_{i,t}^{\text{wm}} + \text{PV}_{i,t}$$

CityLearn's own bookkeeping

$$\text{cost}_{i,t} = e_{i,t} v_t \quad (\text{can be negative})$$

$$\text{emission}_{i,t} = \max(0, e_{i,t} \kappa_t)$$

Our KPIs

$$\text{cost} = \sum_{t,i} \max(e_{i,t}, 0) v_t \quad (\text{imports only})$$

$$\text{emission} = \sum_{t,i} \max(e_{i,t}, 0) \kappa_t$$

$E^{\text{cool}}, E^{\text{heat}}, E^{\text{dhw}}$: device electricity; E^{nsf} : non-shiftable load; E^{batt} : battery (charge +, discharge -);
 E^{EV} : chargers; E^{wm} : washing machines (none here); v_t : price; κ_t : carbon intensity.

How indoor temperature is simulated

LSTMDynamicsBuilding: every house has its own trained LSTM surrogate.

- 2 layers, hidden size 8, look-back 13 h, one .pth file per house.
- **13 min-max normalised inputs:** sin / cos of month, day type and hour; cooling demand; heating demand; direct and diffuse irradiance; outdoor temperature; occupant count; and the **previous indoor temperature**.
- Output: indoor dry-bulb temperature for the current step, un-normalised and written back into the state.
- Warm-up: the heat pump is only controllable once the look-back is filled.

Why it matters

Comfort responds to HVAC actions through a **learned model**, not EnergyPlus itself. Comfort results are only as good as that surrogate.

Consequence for us

Indoor temperature is a **simulated state**, not a data column, whenever the heat pump is controlled.

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Structure of the observation vector

For each building i :

$$o_i = \left[\underbrace{x_i^{\text{base}}}_{28}, \underbrace{x_i^{\text{hp}}}_{2 \text{ (heat-pump mode)}}, \underbrace{x_{i,1}^{\text{ev}}, \dots, x_{i,K}^{\text{ev}}}_{5 \text{ per charger slot}} \right]$$

Environment	dimension
Travis 8 houses	28
Travis, heat-pump mode	30
Travis + EVs (merged)	35
EV challenge, 17 houses	38

Selected by name

Features are mapped from CityLearn's names, not positions, so the selection is auditable and survives schema reordering. A missing feature raises instead of being zero-filled.

The 30 base states (1/2)

#	Feature	Meaning, unit	Where it comes from
0	day_type	day of week, 1 = Monday ... 7 = Sunday	building CSV (calendar)
1	hour	hour of day, 1–24	building CSV (calendar)
2	outdoor_dry_bulb_temperature	°C	weather.csv (actual meteorological year)
3–5	_predicted_1,2,3	forecast at +6, +12, +24 h	weather.csv (verified by lag analysis)
6	diffuse_solar_irradiance	W/m ²	weather.csv
7–9	_predicted_1,2,3	+6, +12, +24 h	weather.csv
10	direct_solar_irradiance	W/m ²	weather.csv
11–13	_predicted_1,2,3	+6, +12, +24 h	weather.csv
14	carbon_intensity	kg CO ₂ /kWh	carbon_intensity.csv (2022 challenge)
15	indoor_dry_bulb_temperature	°C	simulated: per-house LSTM
16	non_shiftable_load	kWh	building CSV (ResStock end use)
17	solar_generation	kWh	PV size × PySAM per-kW series

The 30 base states (2/2)

#	Feature	Meaning, unit	Where it comes from
18	dhw_storage_soc	tank state of charge $\in [0, 1]$	device state (tank model)
19	electrical_storage_soc	battery state of charge $\in [0, 1]$	device state (battery model)
20	net_electricity_consumption	kWh (negative = export)	computed each step (net load formula)
21	electricity_pricing	\$/kWh	pricing.csv (2022 challenge)
22-23	_predicted_1,2	+6, +12 h, exact	pricing.csv (perfect foresight)
24	cooling_demand	kWh thermal	building CSV (EnergyPlus load)
25	dhw_demand	kWh thermal, current hour only	building CSV
26	occupant_count	persons	building CSV
27	indoor_...cooling_set_point	°C	building CSV
28	indoor_...heating_set_point	°C (heat-pump mode)	building CSV
29	heating_electricity_consumption	kWh (heat-pump mode)	computed : heat-pump draw

heating_demand exists in the building CSV but is not an active observation, so the heating electricity draw is used as the heating-load signal.

Five kinds of state, five different origins

Exogenous data files

weather, forecasts, prices, carbon, calendar, loads, demands, occupancy, set points

Learned simulator state

indoor temperature (per-house LSTM)

Device states

battery SOC, tank SOC, EV SOC

Computed aggregates

net consumption, device electricity, PV generation

Declared plans

EV connection, departure countdown, required SOC, battery size (synthetic)

Estimated by us

future hot-water demand (online forecaster)

Only the actions change device states, computed aggregates and the LSTM temperature. Everything in the blue box is fixed data: the agent cannot influence weather, prices or demand, only when energy is drawn and stored.

Calendar conventions: verified against real dates

Hour

step	0	1	2	3
hour	1	2	3	4
step	22	23	24	25
hour	23	24	1	2

$\Rightarrow$ hour $\in \{1, \dots, 24\}$, where 1 is 00:00–01:00.

Day type

step	date (2018)	day_type
0	1 January, Monday	1
672	29 January, Monday	1
4344	1 July, Sunday	7

$\Rightarrow$ 1 = Monday ... 7 = Sunday.

Two defects found while preparing this presentation

- Our hot-water forecaster treats $\text{day_type} \in \{1, 7, 8\}$ as weekend: **Monday** counted as weekend, **Saturday** as weekday.
- It clips hour to 0–23: hours 23 and 24 share a slot and slot 0 never fills.

Both are **to fix**; the pre-heating results (E4, E5) use the forecaster and will be recomputed.

How the forecasts are generated: a lag analysis of the data

For each `_predicted_k` column we searched the lag $\ell \in \{0, \dots, 48\}$ that minimises $\frac{1}{T} \sum_t |\hat{x}_t^{(k)} - x_{t+\ell}|$.

Variable	predicted_1	predicted_2	predicted_3	signal mean abs. dev.
Outdoor temperature	+6 h, MAE 0.15 °C	+12 h, 0.33	+24 h, 0.69	7.59 °C
Direct irradiance	+6 h, 2.69 W/m ²	+12 h, 5.36	+24 h, 10.68	273.5 W/m ²
Diffuse irradiance	+6 h, 0.81 W/m ²	+12 h, 1.60	+24 h, 3.23	70.4 W/m ²
Relative humidity	+6 h, 0.83 %	+12 h, 1.65	+24 h, 3.28	18.1 %
Electricity price	+6 h, 0.0000	+12 h, 0.0000	+24 h, 0.0000	0.095 \$/kWh

Reading

Weather forecasts are the future actuals plus **small noise that grows with horizon**. Price forecasts are **perfect foresight**.

Runtime noise

CityLearn can add Gaussian noise (`noise_std`) to weather actuals and forecasts; its default is 0, so nothing is added at runtime.

What the forecast horizons mean for control

A third defect

Our thermal code assumed the three predictions were **+1, +2, +3 h**. They are **+6, +12, +24 h**. Affected, and **to fix**:

- the temperature uplift in the hot-water forecaster,
- the cold-front term (it looks 6–24 h ahead, not 1–3 h),
- the stress-test front, which ramped per forecast, not per hour.

Horizons against lead times

Hot-water tank fill time	1.6 h
EV fill time	5.7–7.4 h
Weather / price forecasts	6, 12, 24 h

Implications

- Forecasts reach beyond the tank's lead time and cover the EV's.
- Perfect price foresight makes price-aware results **optimistic**: the forecast-error experiment is needed.

Borrowed data

The Travis schema points to `pricing.csv` and `carbon_intensity.csv` from the **CityLearn 2022 challenge (California)**. They are not a Texas tariff.

Price levels	0.21, 0.22, 0.40, 0.50, 0.54 \$/kWh
Structure	time-of-use
Carbon intensity	0.070–0.282 kg CO ₂ /kWh

How we use them

- Economic reward term: $-\mu v_t e_{i,t}$.
- Cost and emission KPIs (imports only).
- Price forecasts feed the policy (+6, +12 h).

Caveat for claims

Any cost number is relative to this tariff; the Thermia and Nordic scenarios need their own spot prices.

What the agent cannot observe, and how we handle it

Missing information	Why it matters	What we do
Future hot-water demand	Tank needs 1.6 h to fill; demand is only visible in the current hour	Causal online climatology per house, hour and day type (next slide)
Heating demand Future EV arrivals	Not an active observation Deadlines appear only on arrival	Heating electricity draw as proxy Barrier acts on connected vehicles; incoming-vehicle signals exist but are not in the STEMS subset
Tank temperature	Legionella needs 60 °C; CityLearn tanks store energy only	Temperature proxy or tank extension (planned)
Forecast uncertainty	Data forecasts are near-perfect	Controlled error injection (planned experiment)

Estimating future hot-water demand (causal, online)

For house i , hour slot h and day class w (weekday or weekend), an exponentially weighted mean over **already observed** steps:

$$m_{i,h,w} \leftarrow m_{i,h,w} + \alpha (d_{i,t} - m_{i,h,w}), \quad \alpha = 0.2$$

Forecast over the pre-heat horizon L , with a cold-weather uplift g :

$$\hat{D}_i(L) = \sum_{k=1}^L m_{i, (h+k) \bmod 24, w} (1 + g \operatorname{relu}(\bar{T} - \hat{T}_{t+k})), \quad g = 0.02$$

Properties

- No look-ahead: only past steps enter m .
- Warm-up (first 24 steps): $\hat{D}_i(L) = L \cdot d_{i,t}$, never below current demand.
- Learns on site, per installation, without data leaving the house.

Corrections pending

- Slot: $h = \text{hour} - 1$ (data are 1–24).
- Weekend: $\text{day_type} \in \{6, 7, 8\}$.
- $\hat{T}_{t+k}$ must use the forecast whose horizon matches (6/12/24 h).

EV states and how the schedules are generated

Per charger slot (5 states)

connected_state	1 if a car is plugged in
departure_time	countdown in steps
required_soc_departure	fraction (CSV stores %)
soc	current state of charge
battery_capacity	kWh of the car present

Hardware (6 of 8 houses): chargers alternate 11 / 7.4 kW, $\eta = 0.95$, 7.2 kW discharge (V2G capable); batteries 60, 40, 75, 52, 60, 40 kWh.

Synthetic schedules (ResStock has no vehicles)

	weekday	weekend
stays home	12 %	45 %
departs	$\mathcal{N}(7.5, 0.8)$ h	$\mathcal{N}(9.5, 1.5)$ h
returns	$\mathcal{N}(17.5, 1.2)$ h	$\mathcal{N}(15.5, 2.0)$ h

- Required SOC $\sim \mathcal{U}(0.70, 0.90)$.
- Arrival SOC
 $\sim \mathcal{U}(0.25, 0.55) - 0.01(\text{hours away} - 8)$,
clipped to $[0.10, 0.60]$.
- Countdown reaches 0 **while still parked**: the departure hour is a last charging opportunity.

Heterogeneous houses: one shape for the graph network

The problem. In the EV challenge dataset houses differ: native actions 1–3, observations 28–42, charger ids embedded in feature names. The GCN stacks houses into one matrix, so every house needs the same vector.

Canonical padded layout

- States = base features + 5K EV slots; a house with fewer bays is zero-padded.
- Actions = canonical order restricted to devices some house owns.
- Per-house index map; presence mask $M \in \{0, 1\}^{B \times A}$.

Two design choices

- A padded slot reports `connected_state = 0`, which the EV barrier already reads as “no vehicle”: one code path, no special case.
- Actions for absent devices are **dropped** before CityLearn, not sent as zeros.

Verified

17 heterogeneous houses $\rightarrow$ one (38,) vector; on Travis every map is the identity, so earlier results are unchanged.

Running normalisation

Each feature j is standardised with a running mean and variance updated from every training episode:

$$\hat{o}_{i,j} = \frac{o_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

The normaliser is saved with the model and reused unchanged at evaluation.

History window

Each house keeps its last $T = 24$ hourly observations,

$$H_{i,t} = [\hat{o}_{i,t-23}, \dots, \hat{o}_{i,t}] \in \mathbb{R}^{24 \times d},$$

which the temporal encoder attends over. The spatial encoder sees the current $\hat{o}_{i,t}$ of all houses.

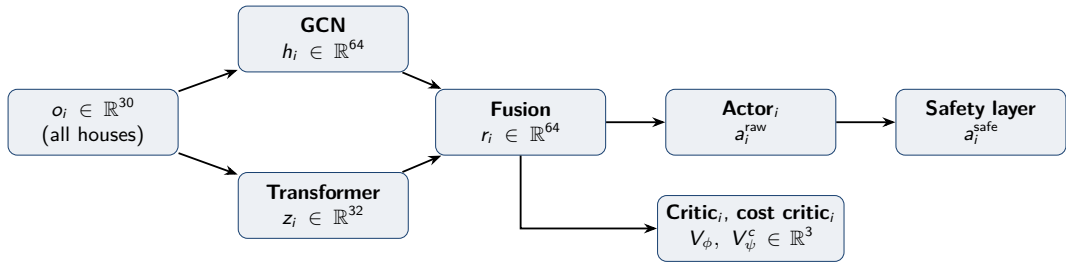
Why both

Weather, prices and demand have daily structure (temporal); houses on one feeder share weather and constraints (spatial).

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From states to actions



The encoder is **shared** by all houses; each house has its **own** actor, value critic and three-headed cost critic (battery SOC, house power, grid power).

Building-similarity graph

$$w_{ij} = \alpha \exp\left(-\frac{d_{ij}^2}{2\sigma_d^2}\right) + \beta \exp\left(-\frac{\|f_i - f_j\|^2}{2\sigma_f^2}\right), \quad w_{ii} = 0$$

- d_{ij} : geographic distance (latitude, longitude).
- f_i : normalised battery capacity and floor area.
- $\alpha = \beta = 0.5$, $\sigma_d = \sigma_f = 1$.

Finding

On the eight co-located Travis houses every off-diagonal weight is **0.80–1.00**. The graph is near-complete and near-uniform, so message passing is close to **averaging over all houses**.

Consequence

Removing the graph (“decentralised”) and keeping it (“coordinated”) moved the raw actions by only 0.001–0.008. A coordination benefit needs either a sharper graph or a constraint that couples the houses.

Spatial encoder: graph convolution

Three layers, width 64, $\sigma = \text{ReLU}$, adjacency with self-loops $\hat{A} = A + I$:

$$H^{(l+1)} = \sigma\left(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad \hat{D}_{ii} = \sum_j \hat{A}_{ij}, \quad H^{(0)} = [\hat{o}_1, \dots, \hat{o}_B]^\top$$

Reading

Each house's representation mixes its own state with its neighbours', weighted by similarity.

The decentralised ablation in one line

Set $A = 0$. Then $\hat{A} = I$ and h_i depends only on house i : same parameters, same optimiser, one knob.

Temporal encoder: self-attention over 24 hours

Four heads, 32-dimensional embedding, fixed sinusoidal positional encodings, window $T = 24$:

$$e_{\tau, \tau'} = \frac{Q_{\tau} K_{\tau'}^{\top}}{\sqrt{d_k}}, \quad z_{i,t} = \text{LayerNorm} \left(x_{i,t} + \sum_{\tau'} \text{softmax}_{\tau'}(e_{t, \tau'}) V_{\tau'} \right)$$

with Q, K, V linear projections of the embedded history $H_{i,t}$.

What it can learn

Daily patterns in load, prices and occupancy, and which past hours predict the next decision, without a hand-written feature for each.

$$r_i = W_s h_i + W_t z_i + b \in \mathbb{R}^{64}$$

Squashed-Gaussian actor (per house)

$$u \sim \mathcal{N}(\mu_\theta(r_i), \sigma_\theta(r_i)^2), \quad a = \tanh(u)$$

$$\log \pi_\theta(a \mid r_i) = \log \mathcal{N}(u; \mu, \sigma^2)$$

$$- \sum_d \log (1 - \tanh^2(u_d) + \epsilon)$$

Critics (per house)

$$V_\phi(r_i) \in \mathbb{R} \quad \text{value}$$

$$V_\psi^c(r_i) \in \mathbb{R}^3 \quad \text{expected cost of}$$

SOC, house power, grid power

Hidden width 128, learning rate $3 \cdot 10^{-4}$, discount $\gamma = 0.99$, target critics with soft update $\tau = 0.005$, entropy temperature auto-tuned to target $-|\mathcal{A}|$.

Reward: four terms

$$r_i^{\text{econ}} = -\mu v_t e_i$$

$$r_i^{\text{stab}} = \alpha_g \left(1 - \frac{E_{\text{grid}}}{P_{\text{grid,max}}}\right)^2 + \alpha_b \left(1 - \frac{|e_i|}{P_{\text{bldg,max}}}\right) - \beta_r \frac{|e_i - e_i^{\text{prev}}|}{P_{\text{bldg,max}}}$$

$$r_i^{\text{comfort}} = -\lambda_c d(T_i^{\text{in}})^2, \quad d(T) = \begin{cases} T - T^{\text{cool}} & T > T^{\text{cool}} \\ T^{\text{heat}} - T & T < T^{\text{heat}} \\ 0 & \text{inside the deadband} \end{cases}$$

$$r_i^{\text{renew}} = \xi \min \left(\frac{\text{solar}_i}{\max(\text{load}_i, \epsilon)}, 1 \right)$$

$E_{\text{grid}} = \sum_i \max(e_i, 0)$. Weights: $\mu = 1$, $\alpha_g = 0.5$, $\alpha_b = 0.3$, $\beta_r = 0.2$, $\lambda_c = 0.4$, $\xi = 0.6$.

Correction over the paper

The paper's single cooling set point penalises comfortable winter temperatures; the dual set-point deadband is used whenever the heat pump is controlled.

Reward: EV service term, and why its weight is what it is

$$r_i^{\text{EV}} = -w_s \mathbb{I}[\text{departs}] (s_i^{\text{req}} - s_i)^+ - w_{sh} \mathbb{I}[\text{connected}] (s_i^{\text{req}} - s_i)^+$$

Defect that motivated it

The original reward had **no EV term**: charging only ever cost money, so “never charge” was optimal. With that reward and an optimistic charge rate, **97.4%** of departures were missed; after adding the term and derating the rate, **2.5%**.

Weight derived from a cost comparison

Filling a 0.5 SOC gap on a 60 kWh car draws $\approx 30 \text{ kWh} \approx 7.5 \$$ at $0.25 \$/\text{kWh}$. The penalty must exceed it: $0.5 w_s > 7.5 \Rightarrow w_s > 15$. We use $w_s = 25$, $w_{sh} = 0.5$.

What happened, honestly

Raising w_s from 2 to 25 moved missed departures only from 100% to about 97% without the barrier. The bottleneck is **credit assignment and training budget**, not the weight.

Constrained actor-critic update

Value and cost targets, generalised advantage estimation ($\gamma = 0.99$, $\lambda = 0.95$):

$$\begin{aligned}y_t &= r_t + \gamma V_\phi^{\text{tgt}}(r_{t+1}), & y_{t,k}^c &= c_{t,k} + \gamma V_{\psi,k}^{c,\text{tgt}}(r_{t+1}) \\ \delta_t &= r_t + \gamma V_\phi(r_{t+1}) - V_\phi(r_t), & \hat{A}_t &= \sum_{l \geq 0} (\gamma \lambda)^l \delta_{t+l} \quad (\text{then standardised})\end{aligned}$$

Constraint costs from the **observed** next state, $k \in \{\text{SOC, house power, grid power}\}$:

$$c_{t,k} = \mathbb{1}[\text{constraint } k \text{ violated at } t+1]$$

Effective advantage used by the actor:

$$\hat{A}_t^{\text{eff}} = \hat{A}_t - \sum_k \max(\lambda_k, 0) (y_{t,k}^c - V_{\psi,k}^c(r_t))$$

Lagrange multipliers with a PID controller

For each constraint k , with measured cost rate J_{c_k} and target rate $d = 0.05$, error $\epsilon_k = J_{c_k} - d$:

$$I_k \leftarrow \text{clip}(I_k + k_i \epsilon_k, 0, \lambda_{\max})$$
$$\lambda_k \leftarrow \text{clip}\left(k_p \epsilon_k + I_k + k_d \text{relu}(J_{c_k}^{(t)} - J_{c_k}^{(t-1)}), 0, \lambda_{\max}\right)$$

$k_p = 0.05$, $k_i = 0.005$, $k_d = 0.01$, $\lambda_{\text{init}} = 0.1$, $\lambda_{\max} = 10$. With $k_p = k_d = 0$ this reduces to plain dual ascent.

Why PID

Integral-only dual ascent oscillates and overshoots; the proportional and derivative terms damp it.

Division of labour

The **shield** enforces the hard limit now; the **Lagrangian** teaches the policy to need the shield less.

Fixing the training scheme so it actually learns

Before

One full-batch gradient step per episode — very few updates in a run. Reward flat at 443 ± 4 ; the policy mean never left its initialisation.

After: $K = 4$ epochs of minibatches of 512 steps, with a clipped surrogate (safe for repeated passes over the same data):

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | r_t)}{\pi_{\theta_{\text{old}}}(a_t | r_t)}$$

$$\mathcal{L}_{\text{actor}} = -\mathbb{E} \left[\min \left(\rho_t \hat{A}_t^{\text{eff}}, \text{clip}(\rho_t, 1 \pm \varepsilon) \hat{A}_t^{\text{eff}} \right) \right] \\ + \alpha \mathbb{E} \left[\log \pi_\theta(\tilde{a}) \right]$$

$\varepsilon = 0.2$; $\tilde{a} \sim \pi_\theta$ is a **fresh sample**.

Two more corrections

- Entropy measured on a fresh policy sample, not on the shielded action.
- Actor target selectable: safe (executed action, paper) or raw (policy's own sample). A deadline barrier outputs exactly 0 while charging is not urgent, which pulls safe targets towards 0.

Evidence (barrier off, one seed)

episode	5	6	7	8	9	10
reward	410	426	428	431	465	459
violations	0.077	0.058	0.059	0.048	0.036	0.051

A trend where there was none — not yet a converged result.

Default remains safe, so published heat-pump results stay reproducible.

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Why a shield, and the three classical barriers

Why

A policy can be unsafe during exploration and whenever it meets states it has not learned. A shield projects every proposed action onto a safe set **before** it reaches the device.

Barriers on the predicted next state

$$h_1 : \text{SOC} \in [0.1, 0.9]$$

$$h_2 : |e_i| \leq P_{\text{bldg,max}} = 80 \text{ kW}$$

$$h_3 : \sum_i \max(e_i, 0) \leq P_{\text{grid,max}} = 300 \text{ kW}$$

A fact that shaped everything after

Travis loads are 10–40 kW, so h_2 and h_3 **never bind** at paper values. Only h_1 is active; power mechanisms need tighter caps to be tested at all.

The calibration discovery

Assumed: every battery moves its state of charge by 0.1 per unit action.

Measured: $\rho_i = P_{\text{nom},i}/C_i$ from each real

battery:	house	1	2	3	4
	ρ_i	0.267	0.244	0.500	0.515
	house	5	6	7	8
	ρ_i	0.244	0.533	0.178	0.207

Consequence

The assumed rate under-estimates the real move by **2–5 times**. “Safe” actions overshoot the band: 9.45% violations **with the shield on**.

Lesson

The single most consequential modelling choice was a constant nobody had checked against the simulator.

Analytic, feasibility-guaranteed projection

Robust margin $m = 0.03$, anticipatory horizon $H = 1$, per-house rate ρ_i :

$$[\ell_i, u_i] = \left[\text{SOC}_{\min} + m + \frac{1}{2}\rho_i H, \text{SOC}_{\max} - m - \frac{1}{2}\rho_i H \right]$$

$$a_i^{\text{batt, safe}} = \begin{cases} \text{clip}\left(a_i^{\text{nom}}, \frac{\ell_i - \text{SOC}_i}{\rho_i}, \frac{u_i - \text{SOC}_i}{\rho_i}\right) & \text{feasible} \\ +1 & \text{SOC}_i < \ell_i \quad \text{recover: charge} \\ -1 & \text{SOC}_i > u_i \quad \text{recover: discharge} \end{cases}$$

Why analytic

h_1 is decoupled across houses: an exact closed form per house, fast enough for 8760×8 steps, no solver.

Why never zeros

A recovery action always exists, even outside the band; the old solver-based shield returned zeros when infeasible.

After the state-of-charge projection, with power derating $\delta = 0.05$:

$$\hat{e}_i = e_i + \max(a_i^{\text{batt}}, 0) P_{\text{nom},i} \quad (\text{charging adds import})$$

$$\hat{e}_i > (1 - \delta)P_{\text{bldg},\text{max}} \Rightarrow a_i^{\text{batt}} \leftarrow \max\left(\text{recovery floor}, \frac{(1 - \delta)P_{\text{bldg},\text{max}} - e_i}{P_{\text{nom},i}}\right)$$

$$\sum_i \max(\hat{e}_i, 0) > (1 - \delta)P_{\text{grid},\text{max}} \Rightarrow a_i^{\text{batt}} \leftarrow a_i^{\text{batt}} \cdot \frac{(1 - \delta)P_{\text{grid},\text{max}}}{\sum_i \max(\hat{e}_i, 0)} \text{ for charging houses}$$

Weather-aware extension (heat pump)

Heat-pump draw is counted, and the cap is derated when the CoP is below its rated value (two slides ahead).

The unification: three devices are one object

A store that must reach a required state of charge by a deadline, charging at a finite rate.

	Hot-water tank	Electric vehicle	Battery
Deadline	implicit: forecast horizon L	declared: departure countdown	none (a band)
Requirement	forecast demand / capacity	declared: required SOC	$\text{SOC}_{\min}$
Capacity	fixed	varies per arrival	fixed
Rate ρ	$\min(c, P\eta/C)$	$0.85 P_{\text{ch}}\eta/C_{\text{car}}$	P_{nom}/C
Measured lead time	1.2–1.9 h	5.7–7.4 h	–
Slack	always 0	positive	–

Why unify

One implementation (DeadlineStorageBarrier), one set of tests, and a single place where the horizon rule lives. Hot water and EVs differ only in their requirement function.

The horizon rule and the monotone projection

With gap $g_i = s_i^{\text{req}} - s_i$ and rate ρ_i :

$$\text{steps_needed}_i = \left\lceil \frac{g_i}{\rho_i} \right\rceil, \quad \text{slack}_i = \text{steps_to_deadline}_i - \text{steps_needed}_i$$

$\text{slack}_i \leq 0$ is a **latest-start trigger**: charge now or miss. Then

$$a_i^{\text{safe}} = \max \left(a_i^{\text{nom}}, \min \left(c_i, \frac{g_i}{\rho_i} c_i \right) \right)$$

Properties

- **Monotone**: raises the action, never lowers it.
- **Feasible**: the requirement is capped below a full store.
- Hot water passes $\text{steps_to_deadline} = 0$: urgent on sight, reproducing the original projection.

Idea behind it

A horizon shorter than the fill time cannot be safe. Choose it from the **measured** lead time, or let the data declare it (EVs).

$$h_4 : \text{SOC}_i^{\text{dhw}} \geq s_i^{\text{req}} = \text{clip}\left(\frac{\hat{D}_i(L)}{C_i} + m_{\text{dhw}} + \kappa \delta_i^{\text{cop}}, 0, 0.95\right)$$

$L = 2$ h, margin $m_{\text{dhw}} = 0.05$, weather gain $\kappa = 0.15$.

Three parts

- forecast demand over L hours,
- robust margin,
- extra energy ahead of a cold front.

Scored by a fixed scorer

All configurations are graded by the **same** readiness barrier ($L = 2$, no weather term), separate from the one being varied, so a longer horizon is not penalised for asking more.

CoP-aware power cap

$$P_{\max,i}^{\text{eff}} = P_{\max} \left(1 - \text{clip} \left(1 - \frac{\text{CoP}_i(T_{\text{out}})}{\text{CoP}_i(T_{\text{ref}})}, 0, 0.5 \right) \right)$$

Heat-pump draw $|a|P_{\text{nom}}$ is counted in the predicted import; the cap tightens when the heat pump is least efficient.

Cold-front anticipation

$$\delta_i^{\text{cop}} = \text{clip} \left(\frac{\text{CoP}_i(T_t) - \min_k \text{CoP}_i(\hat{T}_{t+k})}{\text{CoP}_i(T_t)}, 0, 1 \right)$$

Zero under steady or warming weather; grows when a forecast cold spell will make heating expensive.

Horizon correction

$\hat{T}_{t+k}$ are the data's +6, +12 and +24 h forecasts, not +1, +2, +3 h as originally assumed. The mechanism anticipates a front 6–24 h away; code and documentation are **to fix**.

EV barrier: declared deadlines, measured rates

$$s_i^{\text{req}} = \text{required SOC}, \quad \text{steps_to_deadline}_i = \text{departure countdown}, \quad \rho_i = 0.85 \frac{P_{\text{ch},i} \eta_i}{C_{\text{car},i}}$$

Why derate by 0.85

Nameplate $P\eta/C = 0.174$ against a measured gain of 0.156–0.160. A latest-start trigger fed an optimistic rate starts too late and misses by a hair every time.

Encoding defects we fixed

The countdown was first read as a clock hour, and the generator stopped one step early.

Actuator deadband

Minimum charging power 1.4 kW on an 11 kW charger: **every action in $[0, 0.127]$ gives the**

	action	SOC gain
same charge.	0.03, 0.06, 0.10	+0.0177
	0.20	+0.0293
	1.00	+0.1562

A policy whose outputs sit in that band cannot influence the environment at all.

Where the guarantee breaks: coupled feasibility

Each barrier is feasible on its own. Under a **shared** cap, meeting every deadline requires

$$\sum_i \frac{(s_i^{\text{req}} - s_i) C_i}{\eta_i} \leq P_{\text{cap}} \Delta t T$$

with T the steps to the earliest binding deadline. When it fails, the **safe set is empty**: no sequence of actions meets every deadline and the cap.

Contrast

A battery band is decoupled and always recoverable. Six deadline-locked cars behind one transformer are not.

What the implementation does

It **detects** infeasibility, **degrades** by an explicit priority rule, and **reports** which requirement slips, instead of claiming a guarantee that does not hold.

Sharing a cap: three allocation rules

Available headroom $H_t = P_{\text{cap}} - \sum_i \max(e_i, 0)$; requested charging power $r_i = a_i P_{\text{ch},i}$; total $R = \sum_i r_i$.

Independent

Each barrier projects alone.
Correct per device; blind to the connection.

Proportional

If $R > H_t$:

$$a_i \leftarrow a_i \frac{H_t}{R}$$

Earliest deadline first

Sort by deadline; grant

$$g_i = \min \left(r_i, H_t - \sum_{j \prec i} g_j \right)$$

A design error we made first

We first triggered coordination on energy feasibility (the boxed inequality). Overnight there is almost always enough **energy**; what binds is **instantaneous power**. Proportional and EDF enforce the same total, so their difference measures the priority rule alone.

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Table I metrics (from the observed state)

$$\begin{aligned}\text{cost} &= \sum_{t,i} \max(e_{i,t}, 0) v_t & \text{emission} &= \sum_{t,i} \max(e_{i,t}, 0) \kappa_t \\ \text{avg. daily peak} &= \frac{1}{D} \sum_d \max_{t \in d} \sum_i \max(e_{i,t}, 0) & \text{consumption} &= \sum_{t,i} \max(e_{i,t}, 0) \\ \text{ramping} &= \frac{1}{T-1} \sum_t \left| \sum_i e_{i,t} - \sum_i e_{i,t-1} \right| & \text{discomfort rate} &= \frac{\#\{\text{occupied steps outside band}\}}{\#\{\text{occupied steps}\}} \\ \text{safety violation rate} &= \frac{1}{TB} \sum_{t,i} \mathbb{1}[h_1 \vee h_2 \vee h_3 \text{ violated}]\end{aligned}$$

Avoidable vs unavoidable

A violation is **unavoidable** if no action in $[-1, 1]$ could have prevented it from the pre-action state. This separates a control failure from a physically forced one.

Extended KPIs (Manal: as many as possible)

PV self-consumption	$\sum (PV - \text{export})^+ / \sum PV$
Self-sufficiency	$\sum (PV - \text{export})^+ / \sum (e + PV)^+$
Peak import, load factor	$\max_t E_t^{\text{grid}}, \overline{E^{\text{grid}}} / \max_t E_t^{\text{grid}}$
Cap exceedance	$\sum_t \max (E_t^{\text{grid}} - P_{\text{grid,max}}, 0) \Delta t \text{ [kWh]}$
Discomfort degree-hours	$\frac{1}{B} \sum_{t,i} (\text{degrees beyond the band}) \cdot \mathbb{1}[\text{occupied}] \Delta t$
Battery equivalent full cycles	$\frac{1}{B} \sum_i \frac{1}{2} \sum_t \text{SOC}_{i,t} - \text{SOC}_{i,t-1} $
HVAC on-transitions	off→on switches of $ a^{\text{hvac}} \geq 0.05$ per house-day (compressor-start proxy)
Barrier intervention	share of steps, and mean size, of $ a^{\text{exec}} - a^{\text{policy}} $ on controlled actuators
Equity	coefficient of variation of cost across houses; worst-house discomfort rate
Hot water	readiness rate, deficit [kWh], demand covered by the tank, mean tank SOC
EV	departures, missed departures and rate, energy shortfall at departure [kWh]

Every KPI is checked by a unit test against hand-computed values; pre-existing Table I values are unchanged.

Every run proves its actuators respond

Storage (battery, hot-water tank)

$$\Delta = \overline{\Delta \text{SOC}}|_{a>0.2} - \overline{\Delta \text{SOC}}|_{a\leq 0.05} > 0.01$$

with at least 20 samples on each side.

Heat pump

$$\max_{\ell \in \{0,1\}} \text{corr}(|a_t^{\text{hvac}}|, E_{t+\ell}^{\text{hvac}}) > 0.3$$

Verdict: verified only if every controlled actuator passes; a run with any failure is excluded; too few commands gives “insufficient”, never a silent pass.

Why

The CityLearn hot-water action was a silent no-op: commanding 0.8 for eight steps left the tank at exactly 0.0. Nothing raised an error.

Tested

A synthetic inert tank fails the check; a responsive one passes.

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E1: battery calibration and safety ablation

Configuration	Violations	Cost
No barrier	0.7195	7676
Barrier, uniform rate 0.1	0.0945	7777
Barrier + real calibration	0.0110	7703
+ robust margins	0.0000	7705
+ anticipatory buffer	0.0000	7662
Full configuration	0.0000	7649

Real CityLearn, 800 steps $\times$ 2 seeds, one replayed nominal policy.

Reading

- Calibration accounts for most of the improvement.
- The full configuration is both safest and cheapest.

Why cost did not rise

Power caps never bind on Travis, so the SOC barrier is the only active constraint, and it is now solved exactly. This will not hold once caps bind.

E2: converged full-year run

Metric	Rule-based	STEMS	ratio
cost	34009	15220	0.448
emission	21072	8689	0.412
avg. daily peak	36.78	16.47	0.448
consumption	142817	55220	0.387
ramping	6.37	4.80	0.754
discomfort	0.0001	0.0001	–
violations	0.6465	0.0000	–

15 full-year episodes, seed 0; reward 205 → 671 with zero violations in **every** episode.

Caveat 1: devices

This run predates the hot-water actuator fix: it is a valid **heat-pump-only** result, not a joint hot-water + heat-pump result.

Caveat 2: the baseline is not comparable

The rule-based agent controls **all three** actuators (STEMS only the thermal ones), and its heat-pump command +0.2, commented “normal cooling”, **heats** in CityLearn.

Plan

Corrected rules (with Manal) and a calibrated MPC; then re-run as the reference configuration over seeds, subsets and periods.

E3: the hot-water lead time

Question: can hot water be produced on demand?

Measurement: per-house charge rate

$$\rho_i^{\text{dhw}} = \min(c_i, P_{\text{nom},i}\eta_i/C_i)$$

- $\rho^{\text{dhw}} \in [0.52, 0.85]$
- $\tau \in [1.17, 1.92]$ h, mean 1.60 h
- validated within 15% against the simulator

Answer

No. A reactive controller is structurally too late: the tank needs more than an hour to refill after demand appears.

Hypothesis it generated

Readiness should improve sharply once the pre-heat horizon exceeds the fill time, i.e. between $L = 1$ and $L = 2$.

E4: anticipatory pre-heating

Configuration	Readiness	Covered	Deficit [kWh]	Cost	\$/kWh hot water
No pre-heat (reactive)	0.907	0.939	1026.5	3021.6	0.2200
+ CoP-aware power guard	0.907	0.939	1026.5	3021.6	0.2200
Pre-heat $L = 1$	0.928	0.950	661.7	3025.9	0.2300
Pre-heat $L = 2$	0.961	0.958	90.6	3033.1	0.2411
Pre-heat $L = 3$	0.989	0.968	33.8	3039.0	0.2936
+ cold-front term ($L = 2$)	0.982	0.964	67.3	3035.6	0.2517
Full ($L = 2$, weather, margin)	0.991	0.970	47.4	3038.5	0.2571

1500 steps (winter) $\times$ 2 seeds; nominal controller = price-aware but reactive; one fixed scorer for all rows.

Status

Computed with the forecaster before the calendar and horizon corrections: numbers **to be recomputed**; the qualitative findings are expected to hold and will be re-checked.

E4: what it showed

Positive 1: physics sets the horizon

Deficit falls **86%** from $L = 1$ to $L = 2$ ($661.7 \rightarrow 90.6$ kWh), then improves slowly, exactly where the 1.6 h fill time predicts.

Positive 2: targeted beats uniform

Full ($L = 2$ + cold front) beats $L = 3$ on readiness and coverage while paying **12% less** per kWh of hot water.

Negative 1: readiness is not free

Cost +0.6%; price paid $0.220 \rightarrow 0.257$ \$/kWh. The barrier **commands** heating at demand-driven hours, which are more expensive than price-chosen ones. Planned: choose the cheapest feasible hour *within* the horizon.

Negative 2: CoP guard inactive

Rows 1 and 2 are identical: loads never approach an 80 kW cap. Planned: test under binding caps and in a heating-dominated climate.

Seed spread was exactly 0: determinism, not robustness.

E5: cold-snap stress, after correcting our own design twice

Design corrections

- 1 A constant temperature offset cannot exercise a cold-**front** term (it measures a change) $\rightarrow$ forecast-only ramp.
- 2 A 12 kW cap is not binding when house draw is 0.88 kW $\rightarrow$ 2 kW cap.

Both act on observations, not on the simulator's physics: the offset shifts the observed temperature *and* its three forecasts, the ramp adds $-4, -8, -12^\circ\text{C}$ to them. Intended as -4°C/h , but those forecasts are 6, 12 and 24 h ahead, so the real front is far gentler.

Configuration	Ready	Deficit	Cost	Viol.
No pre-heat	0.895	1163.5	3021.6	0.2402
+ CoP guard	0.895	1163.5	3008.2	0.2388
$L = 1$	0.919	734.7	3013.1	0.2390
$L = 2$	0.961	103.2	3021.4	0.2390
$L = 3$	0.991	33.2	3028.9	0.2402
+ cold front	0.986	60.2	3024.9	0.2390
Full	0.991	43.5	3028.3	0.2390

-18°C offset, -4°C per forecast step, 2 kW cap. **To be recomputed** (forecaster corrections).

E5: what it showed

The CoP guard now acts

Cheapest configuration (cost -0.44%): it trims heat-pump draw hardest when efficiency is lowest.

Anticipation pays when there is weather

The cold-front term cuts the deficit 42%; Full matches $L = 3$ readiness at **17% lower** price per kWh (12% under nominal weather).

But it cannot repair the constraint

Violations stay near 24%. With the battery frozen the shield commands only

$$\frac{366.5}{10597.3} = 3.5\% \text{ of consumption.}$$

The other 96.5% is non-shiftable load.

Principle

A shield can only be as effective as the fraction of load it commands.

E6 and E7: EVs on real charger data, and a houses + EV environment

E6: barrier against CityLearn's EV dataset

- 11 kW, $\eta = 0.95$, 60 kWh $\Rightarrow \rho = 0.174$, ≈ 5.7 h to fill.
- 72-step rollout with real arrivals and empty bays: finite, in bounds, inert on empty bays.
- Unreachable departures are **reported**, not absorbed.

E7: why we built a merged environment

	thermal	EVs
Travis 8	full	none
EV challenge	none	yes
Travis + EV (ours)	full	6 of 8

35 states, 4 actions; buildings unchanged; charger schedules referenced, data never copied.

E8: protecting the heat-pump benchmark

Every structural change (barrier unification, heterogeneous houses, merged schema) was checked twice:

- **Operation level:** the old projection kept as a frozen reference and diffed over 200 random states $\times$ 3 horizons $\times$ 2 weather settings.
- **End to end:** the full E4 study re-run and diffed against stored results.

Result, three refactors in a row

Maximum relative difference $7.359 \cdot 10^{-9}$, on average daily peak.

Honest wording

That residual is a one-unit-in-the-last-place float32 effect from summation order. The test asserts agreement to float32 precision, not exact equality.

E9: coupling sweep with a deadline-locked EV fleet

Question: when does coordinating houses matter?

Setting

- 6 cars, 55.2 kW of chargers.
- Non-EV load peaks at 39.6 kW; with all cars charging on demand, 71.5 kW.
- Mean 4.2 cars connected at once.
- Grid cap swept from 80 kW (slack) to 15 kW.

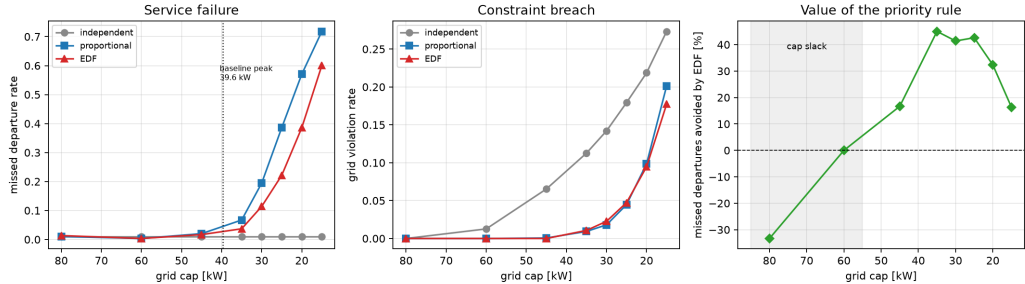
Design

- One uncoordinated nominal controller: each house charges whatever is plugged in.
- Arms: independent, proportional, EDF.
- Service metric: **missed departure** = leaving below required SOC; no later action can repair it.

Why a sweep

All arms coincide when the cap is slack. A single operating point could support almost any conclusion.

E9: result



Left: service failure. Centre: cap breach. Right: missed departures avoided by EDF compared with proportional sharing at the same enforced total. Dotted: 39.6 kW non-EV peak.

E9: the value of coordination is an inverted U

cap kW	independent miss	independent viol	prop. miss	EDF miss	EDF viol	EDF gain
80	3	0.000	3	4	0.000	−33%
60	3	0.013	1	1	0.000	0%
45	3	0.065	6	5	0.000	+17%
35	3	0.113	20	11	0.011	+ 45%
30	3	0.142	58	34	0.023	+41%
25	3	0.179	115	66	0.047	+43%
20	3	0.219	170	115	0.095	+32%
15	3	0.273	214	179	0.177	+16%

298 departures, seed 0, synthetic schedules.

Findings

- Enforcing the cap turns a breach into a scheduling problem (0.113 $\rightarrow$ 0.009 at 35 kW).
- The priority rule is worth **nothing** when slack, **up to 45%** at intermediate caps, and **less again** when most deadlines are unreachable.

Limits

Below the 39.6 kW non-EV peak no charging rule meets the cap. At 80 kW EDF shows 4 misses against 3: the headroom estimate uses the previous import (artifact, fix planned).

Learning experiment: coordinated vs decentralised (attempt)

Design: graph on vs adjacency = 0; shield in independent mode, so coordination must be learned.

First result was invalid

All 8 arms bit-identical across graph, cap **and seed**. Never report a result where the seed does not move.

Four causes found

- ① No EV term in the reward.
- ② Optimistic EV charge rate.
- ③ Actuator deadband $[0, 0.127]$.
- ④ One gradient step per episode.

A finding worth writing up

With the deadline barrier, about 2% of departures were missed; the same training **without** it missed **156 of 156**. A shield strong enough to guarantee an objective **removes the learning signal** for it: the cost-optimal policy is to do nothing.

Status

The training scheme now learns, but EV charging from scratch still misses most departures after short runs. The comparison is **not answered**; it needs a longer budget and an objective the policy, not the shield, is responsible for.

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Defects in the simulator and the data

Defect	Evidence and impact	Status
Hot-water storage action is a no-op (<code>heating_storage.capacity</code> used instead of the tank's)	Action 0.8 for eight steps: tank stays at 0.0. Pre-fix <code>dhw</code> and <code>thermal</code> runs invalid.	fixed (patched loudly)
Shortening the simulation re-sizes every device	Heating heat pump 8.68 kW → 4.34 kW (7-day winter) → 0 kW (7-day summer); CityLearn's demand check fails at step 0.	fixed (episodes; guard raises)
Most candidate houses lack a hot-water tank	61 of 100; our wrapper reported capacity 1.0	to fix
Prices and carbon borrowed from California	Travis schema references 2022-challenge files	documented
Weather forecasts are +6/+12/+24 h; prices exact	Lag analysis of <code>weather.csv</code> and <code>pricing.csv</code>	documented

Defects in our own work (1/2)

Defect	Evidence and impact	Status
Uniform battery rate 0.1	Real 0.178–0.533; 9.45% violations with the shield on	fixed
EV departure countdown read as a clock hour	Wrong deadlines at every bay	fixed
EV countdown truncated one step early	Deleted the last charging opportunity	fixed
SOC metric scored a frozen battery	Thermal isolation reported violation 1.0	fixed
No EV term in the reward	“Never charge” optimal; 97.4% missed, 2.5% after the fixes	fixed
Optimistic EV charge rate	Nameplate 0.174 vs measured 0.156–0.160	fixed (derate)
One gradient step per episode	Reward flat at 443 ± 4	fixed
Coordination gated on energy, not power	EDF identical to independent everywhere	fixed

Defects in our own work (2/2)

Defect	Evidence and impact	Status
Stress test: constant offset and non-binding cap	Could not exercise the mechanisms it was meant to test	fixed (redesigned)
Forecast horizons assumed +1/+2/+3 h	Data are +6/+12/+24 h; affects the forecaster's uplift, the cold-front term and the stress ramp (E4, E5)	to fix
day_type assumed 1 = Sunday	Data: 1 = Monday, so the forecaster counts Monday as weekend and Saturday as weekday (E4, E5)	to fix
hour assumed 0–23	Data: 1–24; forecaster slots 23 and 24 merge and slot 0 never fills (E4, E5)	to fix
Heat-pump mode inferred from the weather	Our CoP guard picks the mode (and its nominal power) from $T_{\text{out}} < \text{set point}$; CityLearn takes it from the sign of the action	to fix
Rule-based baseline misreads both conventions	+0.2 commented “normal cooling” heats; charge/discharge windows shifted by the 1–24 clock. The E2 ratios are against that baseline	to fix
Headroom from previous import	Allocator fires at slack caps (E9, 80 kW)	to fix
Literature summary confabulated by a tool	Claimed a capacity sweep the paper does not contain	fixed (read PDFs)

Why show this list

It tells the audience exactly which numbers were ever at risk, and which results will be recomputed.

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Andreas's seven points and our response

Request	Response
1. Final baseline after the hot-water fix	One reference configuration, all actuators active and verified in every run
2. Proper baselines	Improved rules designed with Manal; perfect-forecast MPC; all calibrated from the simulator
3. Separate RL from the barrier	Five-arm factorial ablation (next slide)
4. Real statistics, forecast errors	Seeds, building subsets, weather periods; 95% CIs and paired differences; error-injection design
5. Binding constraints	Cap sweep from non-binding to strongly binding
6. Heating-dominated scenario	Vermont or Quebec neighbourhoods, re-verified before use
7. Quantitative EV experiment	Deadlines, cost, peaks, devices competing under the same cap sweep

Ablation arms: what each comparison isolates

Arm	Policy	Safety layer
rbc	rules	none
rbc+calibrated	rules	calibrated
rl	learned	none
rl+basic	learned	basic
rl+calibrated	learned	calibrated

basic: uniform rate 0.1, no margins, no anticipation, no hot-water readiness, no CoP guard.

calibrated: measured rates, margins, anticipation, h_4 , CoP guard.

Comparisons

- rl+calibrated – rl: what the safety layer adds to a learned policy.
- rl+calibrated – rbc+calibrated: what learning adds on top of the safety layer.
- rl+calibrated – rl+basic: calibration and anticipation, beyond merely having a barrier.
- rbc+calibrated – rbc: the barrier on a rule-based controller.

The fifth arm (rbc+calibrated) is our addition: without it, policy and barrier cannot be separated factorially.

Sources of variation

- **Training seed:** 3 per learning arm.
- **Building subset:** the reference eight, plus subsets drawn from the **39 fully equipped** houses.
- **Weather period:** per season, train on 28 days and evaluate on the **next** 28 days (unseen weather).

Rule-based arms are deterministic: one run per scenario.

Intervals

$$\bar{x} \pm t_{0.975, n-1} \frac{s}{\sqrt{n}}$$

Paired differences on matched scenarios and seeds:

$$d_j = x_j^{\text{arm}} - x_j^{\text{ref}}, \quad \bar{d} \pm t_{0.975, n-1} \frac{s_d}{\sqrt{n}}$$

Why paired

Differences between scenarios dominate raw spreads; pairing removes them.

Periods as episodes, never as a shorter simulation

Construction	Heat pump, house 134795	Runs?
Full year	8.68 kW	yes
7-day winter simulation	4.34 kW	fails
7-day summer simulation	0.00 kW	fails
28-day winter simulation	house 400973: 3.07 vs 4.42	fails
Episode inside full year	8.68 kW	yes

CityLearn autosizes devices from the simulation period.

Solution

Simulate the full year; select periods with `episode_time_steps = [( $t_s$ ,  $t_e$ )]`.

- Identical hardware for every arm, seed and period.
- Resets take 0.1 s (sizing happens once).
- The environment **raises** if a shortened simulation is requested.

Found by the first smoke run of the harness: all ten runs crashed at step 0.

Constraints that bind, colder climate, competing devices

Cap sweep

Grid and house caps from non-binding to strongly binding, reporting violations, cost, comfort, readiness, feasibility and barrier interventions at each level.

Heating-dominated

Vermont or Quebec neighbourhoods. Before use: actuator check, device census, and whether hot water comes from a heat pump (CoP penalty at 60 °C) or a resistive heater.

Competing devices

EVs, heat pump, hot water and battery under one cap: deadline satisfaction, charging cost, grid peaks, and how the shared power is divided.

Lesson carried over

Every result so far where a mechanism looked inactive was measured where it could not act (slack caps, frozen battery, 3.5% of load). The sweep is designed so that each mechanism is tested where it can act.

Forecast-error robustness (Andreas point 4)

Perturbed forecast of variable x at horizon $h \in \{6, 12, 24\}$ h:

$$\tilde{x}_{t+h} = x_{t+h} + b_h + \varepsilon_{t,h}, \quad \varepsilon_{t,h} \sim \mathcal{N}(0, \sigma_h^2), \quad \sigma_h = \sigma_0 \frac{h}{24}$$

Factors

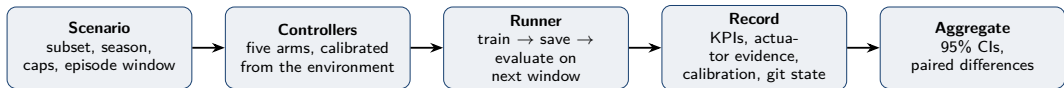
- **Where:** policy inputs / barrier inputs / both.
- **When:** training and evaluation / evaluation only.
- **Type:** zero-mean noise / systematic bias b_h .

CityLearn's `noise_std` is a second, built-in generator for weather.

Hypotheses

- Training with errors makes the **policy** robust (seen in unconstrained DRL).
- **Bias** hurts constraint satisfaction more than noise: under-forecast demand makes the barrier too permissive.
- The barrier's own anticipation degrades with its forecasts; placing errors separately shows which layer carries the risk.

Evaluation harness



Engineering

- Parallel and resumable; failures recorded, not raised.
- Runs whose actuators fail the check are excluded and counted.

Status

- 127 tests passing (library, harness and the new episode-window guard).
- First smoke grid exposed the autosizing defect; next: re-run with episode windows and fully equipped subsets.

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The heat-pump controller runs on its own

Isolation by device name

```
heatpump = [ heat pump ]  
thermal = [ hot water, heat pump ]
```

- Non-controlled devices held at 0: for storage, exactly “do nothing”.
- Battery barrier skipped when the battery is not controlled; power barriers stay on.
- Asking for a device the house lacks raises.

What standalone mode contains

Encoder, policy, hot-water readiness h_4 , CoP-aware power guard, dual set-point comfort. No battery, EV or neighbourhood coupling needed.

Staged adoption

E4 measured h_4 on a conventional reactive controller: an installation can adopt the **safety and anticipation layer first** and a learned policy later.

Integration with a commercial heat-pump platform (Thermia)

Signals needed

Outdoor temperature + forecast	hourly
Indoor temperature, set points	hourly
Tank temperature or state	hourly
Hot-water draw (past)	hourly
Electrical import, price	hourly
Rated power, tank size, efficiencies	once

Steps

- ① **Calibrate at commissioning:** read the installed unit's ratings (the lesson of E1 and E3).
- ② **Barrier at the edge:** analytic, no solver, no neighbourhood state, no connectivity dependence.
- ③ **Forecaster learns on site:** no data leaves the house.
- ④ **Policy optional** at first.

Before any Nordic claim

Heating-dominated climate; ground-source brine temperature instead of outdoor air in the CoP; inverter vs on-off control; hot-water heat pump with CoP penalty at 60 °C.

Legionella as a periodic deadline (proposed)

Thermia rule (via Manal): at least once per week the tank reaches 60 °C for a set duration D (duration being confirmed).

Energy-equivalent state for 60 °C and a counter c_t of hours since the last completed cycle, $W = 168$ h:

$$s^{\text{leg}} = \frac{T^{\text{leg}} - T^{\text{cold}}}{T^{\text{max}} - T^{\text{cold}}}, \quad \text{slack}_t^{\text{leg}} = (W - c_t) - \left\lceil \frac{(s^{\text{leg}} - s_t)^+}{\rho} \right\rceil - D$$

Start the cycle when $\text{slack}_t^{\text{leg}} \leq 0$; before that, the cycle may be moved to the cheapest or warmest feasible hours.

Why it fits

Same object as h_4 and EVs: a required state, a deadline, a finite rate, plus a dwell time D .

Open questions

Cycle length; fixed or movable time; heat pump alone or with an electric backup; CityLearn tanks store energy, not temperature.

Discussed

- Test on different **consumer profiles**: residential and commercial.
- Add **hygiene** (anti-Legionella cycle).
- Report **as many KPIs as possible**.
- Test on **Thermia data** once the model is ready.
- Move the paper **beyond a benchmark study**.

How each maps onto the plan

- Profiles → another source of variation in the statistics.
- Hygiene → a new constraint, not a new mechanism.
- KPIs → extended KPI set (already implemented).
- Thermia → integration steps above.
- Framing → the research question on slide 3.

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RL for domestic hot water (papers shared by Manal)

Paper	Approach	Reported result
Ali and Kazmi (2017), LNCS	Model-free RL, nZEBs, Utrecht (abstract not accessible)	minimise grid interaction of PV and hot water
De Somer et al. (2017), ISGT Europe	Model-based RL schedules hot-water buffer cycles; 6 real homes	self-consumption significantly up vs thermostat
Soares et al. (2020), Energy and Buildings	Model-based RL field test; heat-pump buffers and batteries	+14% PV self-consumption, no discomfort
Heidari et al. (2021), J. Phys. Conf. Ser.	RL balancing energy, comfort and hygiene ; measured use in one home, 8 weeks	24.5% energy saving, comfort and hygiene preserved
Langer and Völling (2022), Applied Energy	DDPG vs full-information MPC and rule-based; heat pump, PV, battery, storage	75% self-sufficiency, limited comfort violations
Amasyali et al. (2023), ORNL	DQN agents, electricity- or emission-focused, on a heat-pump water heater; several usage and marginal-emission profiles	12–22% less electricity; 18–37% lower emissions

Gap (from the five abstracts available)

Comfort and hygiene are objectives or soft penalties; none mentions guaranteed constraint satisfaction, a safety layer, or constraints shared across buildings. Heidari et al. also have two 2022 Applied Energy papers on hygiene, to be read before the Legionella design.

Work	Relation to ours
arXiv:2602.19223 (2026), CityLearn MARL benchmark arXiv:2605.01362 (2026), coordination architectures, Vermont	Decentralised training beats centralised; no analysis of constraint tightness. Our sweep offers an explanation: nothing binds. Hierarchical decomposition works best; explicitly no caps, EVs or deadlines.
arXiv:2607.00935 (2026), deadline-aware EV charging with transformer ageing Input-constrained and backup CBFs (survey)	Soft deadlines and a marginal-cost urgency index (richer than EDF); four discrete scenarios, no capacity sweep. A rule to adopt and compare. Our deadline barrier is a discrete-time, rate-limited instance; the contribution is the measured rate and the horizon rule, not a new theorem.
arXiv:2608.16335 (2026), “readiness barrier functions”	Multicopter actuator authority, different mathematics; name collision avoided.

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Limitations we state openly

- One climate so far (cooling-dominated Texas).
- Prices and carbon borrowed from California; price forecasts exact.
- EV schedules synthetic.
- Seed spreads were zero (determinism), so no confidence intervals yet.
- Heat-pump-only converged result; joint run pending.
- Learned coordination not yet compared (training budget).
- Pre-heating results to be recomputed after the calendar and horizon fixes.
- Controllable share of load small when the battery is frozen.
- Indoor temperature from an LSTM surrogate.
- Tank model has no temperature (Legionella).

Next steps, in order

- ① Fix the forecaster calendar and horizon defects and the tank-capacity fallback; recompute E4 and E5; regression-check.
- ② Re-run the smoke grid with episode windows and fully equipped subsets.
- ③ Reference configuration: five-arm ablation with seeds, subsets and periods; 95% CIs.
- ④ Calibrated baselines: improved rules with Manal; perfect-forecast MPC.
- ⑤ Cap sweep on the joint configuration and all arms.
- ⑥ Heating-dominated scenario; Legionella as a periodic deadline.
- ⑦ Quantitative EV experiment with competing devices; price-aware scheduling within the slack.
- ⑧ Forecast-error robustness; then Thermia data.

- ① **Measure, do not assume:** one unchecked constant explained most violations; one unchecked convention broke the forecaster.
- ② **Match horizons to lead times:** the deficit collapses exactly where the measured fill time says.
- ③ **One abstraction:** hot water, EVs and batteries are stores with deadlines.
- ④ **Coupling changes the guarantee:** under a shared cap the safe set can be empty; detect it and degrade explicitly.
- ⑤ **Coordination has a sweet spot:** worthless when slack, most valuable at intermediate caps.
- ⑥ **Shield and learning interact:** a barrier that guarantees an objective removes the signal to learn it.
- ⑦ **Test mechanisms where they can act:** slack caps and frozen devices hide everything.

Questions and discussion